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Course code and name:	F79BI Bayesian Inference and Computational Methods
Type of assessment:	**Individual**
Coursework Title:	Assignment – Semester 2, 2024-25
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F79BI : ASSIGNMENT

ANN ERINJERI - H00418805

Students with whom I discussed the assignment: Attiya, Fidha, Marwa

Information Given:

The problem is illustrated with the certain quantities, using their respective notations. Table 1 given in the appendix below outlines the quantities against their respective notations.

Tasks:

1. Aim: To compute θ_E, the probability that a seabird is on island E in year 2 and θ_W, the probability that a seabird is on island W in year 2.

   As depicted in Figure 1 (given below), a seabird on island E in year 2 is either

   i. one that was in island E in year 1 (p_E) and did not switch to island W in year 2 (with a probability of 1 – δ), or
   ii. one that was in island W in year 1 ($p_W = 1 - p_E$) but switched to island E in year 2 (with a probability of δ).

   Therefore, $\theta_E = p_E(1-\delta) + (1-p_E)\,\delta$

$$\boldsymbol{\theta_E = p_E(1-2\delta) + \delta}$$

   Consequently, since $\theta_W = 1 - \theta_E$,

$$\boldsymbol{\theta_W = (1-p_E)(1-2\delta) + \delta}$$

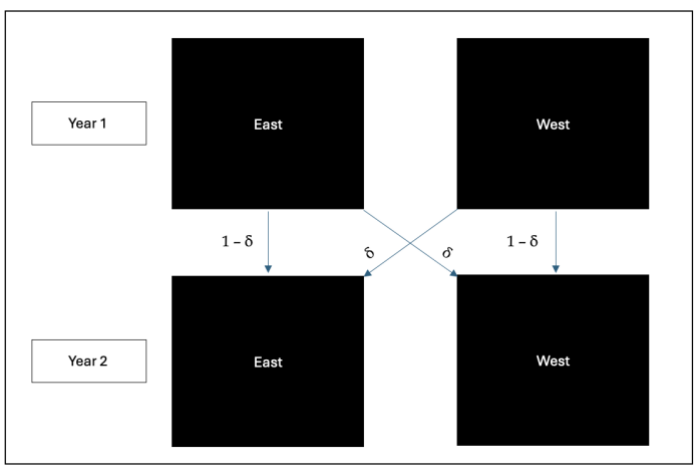

Figure 1: Diagram depicting the migration of seabirds across the islands from Year 1 to Year 2

2. Aim: To compute τ_E, the probability that a seabird is on island E in year 1 given that it is on island E in year 2, and τ_W, the probability that a seabird is on island W in year 1 given that it is on island W in year 2.

   Let us define the following events and their respective probabilities in Table 2.

Event (E)	Probability of Event P(E)
E1: The seabird is on the East Island in Year 1	P(E1) = p_E
E2: The seabird is on the East Island in Year 2	P(E2) = θ_E
W1: The seabird is on the West Island in Year 1	P(W1) = p_W
W2: The seabird is on the West Island in Year 2	P(W2) = θ_W

Table 2 : Events related to the problem and their respective probabilities

As outlined in the problem description, P(E2|E1) = P(W2|W1) = 1−δ and P(E2|W1) = P(W2|E1) = δ

By definition, τ_E = P(E1|E2)

Applying Bayes' Theorem, $\tau_E = P(E2 \mid E1) \cdot P(E1) \,/\, P(E2)$

$$\boldsymbol{\tau_E = \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}}$$

Similarly, τ_W = P(W1|W2)
Applying Bayes' Theorem, $\tau_W = P(W2 \mid W1) \cdot P(W1) \,/\, P(W2)$

$$\boldsymbol{\tau_W = \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}}$$

3. Aim: To compute the marginal distribution of X_E, the number of seabirds on island E in year 2.
   Assumptions: The seabirds move independently of each other.

   There are N number of birds in total, out of which X_E number of seabirds reside on island E in year 2 with probability, θ_E and the rest (N - X_e) will reside on island W in year 2 with probability θ_W.

   Hence, X_E follows a Binomial Distribution [Bin(n,p)] with n = N and p = θ_E.

   Marginal Distribution of X_E : $\mathbf{X_E \sim Binomial(N, \theta_E}$

   PMF of X_E : $\boldsymbol{P(X_E = x_E \mid \theta_E) = \binom{N}{x_E}[\theta_E]^{x_E}[1-\theta_E]^{N-x_E}}$,), with $\boldsymbol{\theta_E = p_E(1-2\delta)+\delta}$

4. Aim: To compute the conditional distributions of Y_E, the number of seabirds on island E for both years 1 and 2 and Y_W, the number of seabirds on island W for both years 1 and 2, conditional to (X_E, p_E, δ).

   Assumptions: Y_E and Yw are independent of each other.

   a) There are X_E number of seabirds on island E in year 2, out of which Y_E number of seabirds were already on island E in year 1 with probability, τ_E and the rest (X_E - Y_E) will have come from island W.

      Conditional Distribution of Y_E conditional to (X_E, p_E, δ **):**

      $Y_E \mid X_E, p_E, \delta \sim$ Binomial(X_E , τ_E), with $\tau_E = \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}$

      $\Rightarrow \mathbf{Y_E \mid X_E, p_E, \delta \sim Binomial(X_E\,, \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta})}$

      PMF of Y_E conditional to (X_E, p_E, δ):

      $$\boldsymbol{P(Y_E = y_E \mid X_E,\ pE,\ \delta) = \binom{X_E}{y_E} \cdot \left(\frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}\right)^{y_E} \cdot \left(1 - \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}\right)^{X_E - y_E}}$$

b) Similarly, there are N - X_E number of seabirds on island W in year 2, out of which Y_W number of seabirds were already on island W in year 1 with probability, τ_W and the rest (N - X_E - Y_W) will have come from island E.

Conditional Distribution of Y_W conditional to (X_E, p_E, δ):

$Y_W \mid X_E, p_E, \delta \sim \text{Binomial}(N - X_E, \tau_W)$, with $\tau_W = \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}$

$\Rightarrow \mathbf{Y_W \mid X_E, p_E, \delta \sim Binomial(N - X_E, \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta})}$

PMF of Y_W conditional to (X_E, p_E, δ):

$$\boldsymbol{P(Y_W = y_W \mid X_E,\ pE, \delta) = \binom{N-X_E}{y_W} \cdot \left(\frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}\right)^{y_W} \cdot \left(1 - \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}\right)^{N-X_E-y_W}}$$

5. Aim: To compute the conditional distribution, π (p_E, δ | x_E, y_E, y_W) of (p_E, δ) given X_E, Y_E and Y_W.
Assumptions: a) $p_E \sim \text{Uniform}(0,1) \Rightarrow \pi(p_E) = 1$, $\delta \sim \text{Beta}(2,10) \Rightarrow \pi(\delta) \propto \delta(1-\delta)^9$

b) Y_E and Yw are independent of each other.

To obtain this conditional distribution, we combine the likelihoods of X_E (observed data), Y_E and Y_W and the priors of p_E and δ.

[Task 3] $P(X_E = x_E \mid p_E, \delta) = \binom{N}{x_E} [\theta_E]^{x_E}[1-\theta_E]^{N-x_E}$

[Task 4] $P(Y_E = y_E \mid X_E,\ pE,\ \delta) = \binom{X_E}{y_E} \cdot \left(\frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}\right)^{y_E} \cdot \left(1 - \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}\right)^{X_E-y_E}$

[Task 4] $P(Y_W = y_W \mid X_E,\ pE, \delta) = \binom{N-X_E}{y_W} \cdot \left(\frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}\right)^{y_W} \cdot \left(1 - \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}\right)^{N-X_E-y_W}$

$\pi(p_E, \delta \mid x_E, y_E, y_W) \propto P(Y_E = y_E \mid X_E,\ pE,\ \delta) \cdot\ P(Y_W = y_W \mid X_E,\ pE, \delta) \cdot P(X_E = x_E \mid p_E, \delta) \cdot \pi(p_E) \cdot \pi(\delta)$

$$\pi(p_E, \delta \mid x_E, y_E, y_W) \propto \left(\frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}\right)^{y_E} \cdot \left(1 - \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}\right)^{X_E-y_E} \cdot \left(\frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}\right)^{y_W} \cdot \left(1 - \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}\right)^{N-X_E-y_W} \cdot \theta_E^{x_E} \cdot (1-\theta_E)^{N-x_E} \cdot 1 \cdot \delta(1-\delta)^9$$

$$\Rightarrow \boldsymbol{\pi(p_E, \delta \mid x_E, y_E, y_W) \propto \tau_E^{y_E} \cdot (1-\tau_E)^{X_E-y_E} \cdot \tau_W^{y_W} \cdot (1-\tau_W)^{N-X_E-y_W} \cdot \theta_E^{x_E} \cdot (1-\theta_E)^{N-x_E} \cdot \delta(1-\delta)^9}$$

with $\tau_E = \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta}$ and $\tau_W = \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta}$

6. Aim: To compute the full conditional distributions π (p_E | x_E, y_E, y_W) and π (δ | x_E, y_E, y_W) of p_E and δ, respectively, given x_E, y_E and y_W.

To compute π (p_E | x_E, y_E, y_W), we modify the conditional distribution,

$$\pi(p_E, \delta \mid x_E, y_E, y_W) \propto \tau_E^{y_E} \cdot (1-\tau_E)^{X_E - y_E} \cdot \tau_W^{y_W} \cdot (1-\tau_W)^{N - X_E - y_W} \cdot \theta_E^{x_E} \cdot (1-\theta_E)^{N - x_E} \cdot \delta(1-\delta)^9$$

We treat δ as a constant. Since x_E is observed, we consider it a constant as well. (We drop constant terms under proportionality)

Hence, $\pi(p_E \mid x_E, y_E, y_W) \propto \tau_E^{y_E} \cdot (1-\tau_E)^{X_E - y_E} \cdot \tau_W^{y_W} \cdot (1-\tau_W)^{N - X_E - y_W}$

Inferences from previous tasks:

$$\tau_E = \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta} \Rrightarrow \tau_E \propto p_E \Longrightarrow (1 - \tau_E) \propto (1-p_E)$$

$$\tau_W = \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta} \Rrightarrow \tau_W \propto (1-p_E) \Longrightarrow (1 - \tau_W) \propto p_E$$

This gives, $\pi(p_E \mid x_E, y_E, y_W) \propto p_E^{N - X_E + Y_E - Y_W} \cdot (1-p_E)^{X_E - Y_E + Y_W}$

$$\boldsymbol{p_E \mid x_E, y_E, y_W \sim \text{Beta}(N - x_E + y_E - y_W + 1,\ x_E - y_E + y_W + 1)}$$

To compute $\pi(\delta \mid x_E, y_E, y_W)$, we modify the conditional distribution,

$$\pi(p_E, \delta \mid x_E, y_E, y_W) \propto \tau_E^{y_E} \cdot (1-\tau_E)^{X_E - y_E} \cdot \tau_W^{y_W} \cdot (1-\tau_W)^{N - X_E - y_W} \cdot \theta_E^{x_E} \cdot (1-\theta_E)^{N - x_E} \cdot \delta(1-\delta)^9$$

We treat p_E as a constant. Since x_E is observed, we consider it a constant as well. (We drop constant terms under proportionality)

Hence, $\pi(\delta \mid x_E, y_E, y_W) \propto \tau_E^{y_E} \cdot (1-\tau_E)^{X_E - y_E} \cdot \tau_W^{y_W} \cdot (1-\tau_W)^{N - X_E - y_W} \cdot \delta(1-\delta)^9$

Inferences from previous tasks:

$$\tau_E = \frac{(1-\delta)p_E}{p_E(1-2\delta)+\delta} \Rrightarrow \tau_E \propto (1-\delta) \Longrightarrow (1 - \tau_E) \propto \delta$$

$$\tau_W = \frac{(1-\delta)(1-p_E)}{(1-p_E)(1-2\delta)+\delta} \Rrightarrow \tau_W \propto (1-\delta) \Longrightarrow (1 - \tau_W) \propto \delta$$

This gives, $\pi(\delta \mid x_E, y_E, y_W) \propto \delta^{N - Y_E - Y_W + 1} \cdot (1-\delta)^{Y_E + Y_W + 9}$

$$\boldsymbol{\delta \mid x_E, y_E, y_W \sim \text{Beta}(N - y_E - y_W + 2,\ y_E + y_W + 10)}$$

7. Aim: To compute an algorithm that uses data augmentation (using the unobserved data y_E and y_W) and Gibbs sampling to investigate $\pi(p_E, \delta \mid x_E)$, the posterior distribution of p_E and δ.

   By definition,

$$\pi(p_E, \delta \mid x_E) \propto [p_E(1-2\delta)+\delta]^{x_E} \cdot [1 - p_E(1-2\delta) - \delta]^{N - x_E} \cdot \delta(1-\delta)^9$$

   This is not tractable, so we use the full conditional distributions obtained in Task 4 and Task 6, and employ Gibbs sampling to investigate the posterior distribution of p_E and δ.

The algorithm used is as given below:

1. Set B to be the desired number of iterations.
2. Set suitable initial values for $p_E \in [0,1]$ and $\delta \in [0,1]$, since p_E is a proportion and δ is a probability value.
3. Sample the unobserved data Y_E from its conditional distribution: Binomial $\left(X_E, ((1-\delta)p_E)/(p_E(1-2\delta)+\delta)\right)$ using the current values (initial values for the first iteration) for p_E and δ.
4. Sample the unobserved data Y_W from its conditional distribution: Binomial $\left(N - X_E, ((1-\delta)(1-p_E))/((1-p_E)(1-2\delta)+\delta)\right)$
   using the current values (initial values for the first iteration) for p_E and δ.
5. Using the sampled values of Y_E and Y_W, sample p_E from its conditional distribution Beta$(N - X_E + Y_E - Y_W + 1, X_E - Y_E + Y_W + 1)$. Update the value of p_E for the next iteration.
6. Using the sampled values of Y_E and Y_W, sample δ from its conditional distribution Beta$(N - Y_E - Y_W + 2, Y_E + Y_W + 10)$. Update the value of δ for the next iteration.
7. Repeat steps 3, 4, 5, and 6 for B number of iterations.

8. Aim: To implement the algorithm described in Task 7 in R, as shown in the appendix.

We can use x_E, the number of seabirds on island E in year 2 over N, the total population of seabirds for our initial guess for p_E, the proportion of seabirds nesting on island E in year 1.

$p_E[1] = x_E / N = 0.704$ [x_E (as per my seed)= 704, N = 1000]

We can use the the mean of the prior distribution of δ, for its initial guess.

$\delta[1] = 2/12 = 0.167$ [Mean of $\delta \sim$ Beta(2,10)]

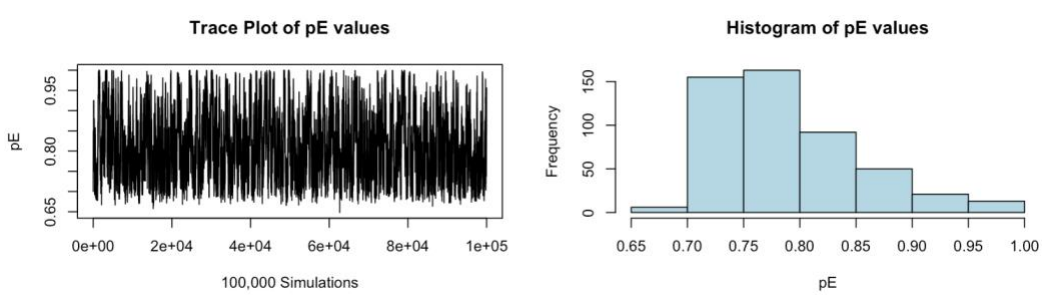

Figure 2 Trace Plot and Histogram of p_E values

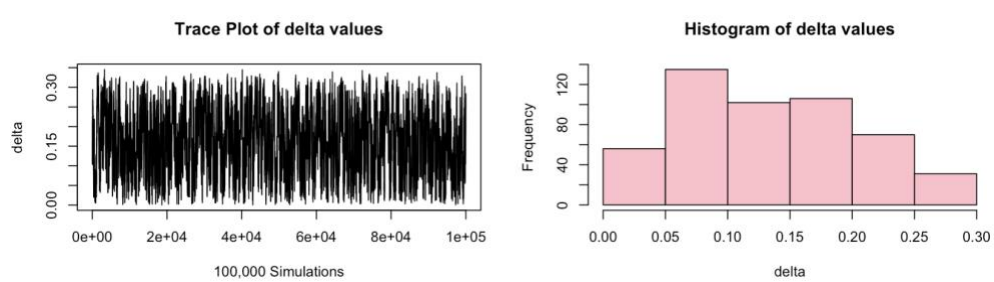

Figure 3 Trace Plot and Histogram of δ values

The trace plots for p_E and δ in Figure 2 and Figure 3, respectively, show satisfactory mixing, showing good fluctuations, ensuring that the chain is not stagnant, and not autocorrelated. The histogram for p_E values in Figure 2 shows strong positive skew. The histogram for δ values in Figure 3 shows slight positive skew.

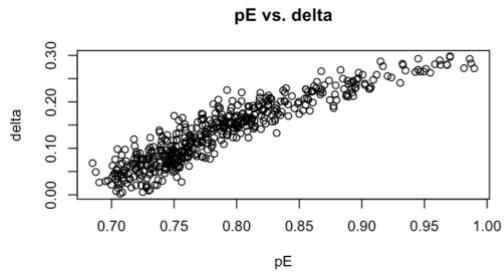

Figure 4 Scatterplot of p_E values against δ values

The scatterplot in Figure 4 shows that p_E values and δ values are strongly positively correlated.

The numerical summaries of p_E values and δ values are given below

Numerical Summary for p_E values

```
> sd(pE[seq(B-(m*n)+1,B,m)])
[1] 0.06408637
> summary(pE[seq(B-(m*n)+1,B,m)])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6849  0.7417  0.7745  0.7883  0.8207  0.9897
```

Numerical Summary for δ values

```
> sd(delta[seq(B-(m*n)+1,B,m)])
[1] 0.06970555
> summary(delta[seq(B-(m*n)+1,B,m)])
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
0.002406 0.077099 0.131324 0.134418 0.185747 0.298395
```

The 95% Confidence Interval for p_E is given by (0.7038834, 0.9503331).

The 95% Confidence Interval for δ is given by (0.02090387, 0.27800788).

Conclusion

The mean of p_E, the proportion of seabirds nesting on island E in year 1, is computed as 0.7883, which is fairly high. The distribution of p_E is highly positively skewed which indicates that majority of the values lie in the range (0.7, 0.85). This is in agreeance with the confidence interval obtained above. The 95% Confidence Interval for p_E is given by (0.7038834, 0.9503331).

The mean of δ, the probability that a seabird will change from W to E, or from E to W, is computed as 0.134418, which is fairly low. The distribution of δ is slightly positively skewed. The 95% Confidence Interval for δ is given by (0.02090387, 0.27800788).

p_E and δ are highly positively correlated, with a correlation constant of 0.9232257.

This shows that there is a high proportion of seabirds nesting on island E in year 1, and a low probability of switching between islands in year 2. The probability of switching increases with the increase in the proportion of seabirds nesting on island E in year 1.

Appendix

1) R code for Task 8

```
set.seed(8805)

N <- 1000 #Total population of seabirds
xE <- sample(400:800, 1) #The number of seabirds on island E in year 2

B <- 100000 #Number of iterations to run #Burn in
n <- 500 # Number of samples
m <- 10 #Thinning

#Initializing vectors for simulation
pE <- vector("numeric", B) #The proportion of seabirds nesting on island E in year 1
delta <- vector("numeric", B) #The probability that a seabird will change from W to E, or from E to W
yE <- vector("numeric", B) #The number of seabirds on island E for both years 1 and 2
yW <- vector("numeric", B) #The number of seabirds on island W for both years 1 and 2

pE[1] <- 0.7 #Initial guess from xE/N
delta[1] <- 0.16 #Initial guess from the prior mean
for (iter in 1:(B - 1)) {

  theta_E = pE[iter] * (1 - 2 * delta[iter]) + delta[iter] #The probability that a seabird is on island E in year 2
  theta_W = 1 - theta_E #The probability that a seabird is on island W in year 2

  tau_E = ((1 - delta[iter]) * pE[iter]) / theta_E #The probability that a seabird is on island E in year 1 given that it is on island E in year 2
  tau_W = ((1 - delta[iter]) * (1 - pE[iter])) / theta_W #The probability that a seabird is on island W in year 1 given that it is on island W in year 2

  #Augmenting the unobserved data yE and yW using their conditional distributions
  yE[iter + 1] <- rbinom(1, xE, tau_E) #The number of seabirds on island E for both years 1 and 2
  yW[iter + 1] <- rbinom(1, N - xE, tau_W) #The number of seabirds on island W for both years 1 and 2

  alpha_pE <- N - xE + yE[iter + 1] - yW[iter + 1] + 1 #The shape parameter for the distribution of pE
  beta_pE <- xE - yE[iter + 1] + yW[iter + 1] + 1 #The rate parameter for the distribution of pE

  alpha_delta <- N - yE[iter + 1] - yW[iter + 1] + 2 #The shape parameter for the distribution of delta
  beta_delta <- yE[iter + 1] + yW[iter + 1] + 10 #The rate parameter for the distribution of delta

  #Updating pE and delta using conditional distributions
  pE[iter + 1] <- rbeta(1, alpha_pE, beta_pE)
  delta[iter + 1] <- rbeta(1, alpha_delta, beta_delta)
}

par(mfrow=c(2,2))

#Plotting Results
```

```
plot(main= "Trace Plot of pE values", xlab="100,000 Simulations",seq(1,B,1), pE , type="l")
hist(main= "Histogram of pE values", xlab = "pE",pE[seq(B-(m*n)+1,B,m)], col = "lightblue")

plot(main= "Trace Plot of delta values", xlab="100,000 Simulations", seq(1,B,1), delta, type="l")
hist(main= "Histogram of delta values",xlab = "delta", delta[seq(B-(m*n)+1,B,m)], col = "pink")

#95% confidence intervals for pE and delta
pE_quantiles <- quantile(pE[seq(B-(m*n)+1,B,m)],probs = c(0.025,0.975))
pE_quantiles
delta_quantiles <- quantile(delta[seq(B-(m*n)+1,B,m)],probs = c(0.025,0.975))
delta_quantiles

#Numerical summaries for pE and delta

sd(pE[seq(B-(m*n)+1,B,m)])
summary(pE[seq(B-(m*n)+1,B,m)])


sd(delta[seq(B-(m*n)+1,B,m)])
summary(delta[seq(B-(m*n)+1,B,m)])

#Checking for correlation between pE and delta
plot(main = "pE vs. delta", xlab="pE", ylab= "delta",pE[seq(B-(m*n)+1,B,m)], delta[seq(B-
(m*n)+1,B,m)])
cor(pE[seq(B-(m*n)+1,B,m)],delta[seq(B-(m*n)+1,B,m)])

#Checking the skewness of both distributions
library(e1071)
skewness(pE[seq(B-(m*n)+1,B,m)])
skewness(delta[seq(B-(m*n)+1,B,m)])
```

2) Table 1

Quantity	Notation
Total population of seabirds	N
The proportion of seabirds nesting on island E in year 1	p_E
The proportion of seabirds nesting on island W in year 1	p_W
The probability that a seabird will change from W to E, or from E to W	δ
The number of seabirds on island E in year 2	x_E
The number of seabirds on island W in year 2	$N - x_E$
The probability that a seabird is on island E in year 2	θ_E
The probability that a seabird is on island W in year 2	θ_W
The probability that a seabird is on island E in year 1 given that it is on island E in year 2	τ_E
The probability that a seabird is on island W in year 1 given that it is on island W in year 2	τ_W
The number of seabirds on island E for both years 1 and 2	Y_E
The number of seabirds on island W for both years 1 and 2	Y_W

Table 1: Information on the notations used for each quantity used to describe the problem